Discussion Forums

Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: cmpe295

Posted on: Mar 21, 2013 8:26 PM

I have configured hadoop on two of my ec2 instances. I am unable to communicate between the two instances. This is the message I see in the slave log -

2013-03-22 02:57:13,476 INFO org.apache.hadoop.ipc.Client: Retrying connect to server: master/54.225.206.51:54310. Already tried 8 time(s).
2013-03-22 02:57:14,478 INFO org.apache.hadoop.ipc.Client: Retrying connect to server: master/54.225.206.51:54310. Already tried 9 time(s).
2013-03-22 02:57:14,481 INFO org.apache.hadoop.ipc.RPC: Server at master/54.225.206.51:54310 not available yet, Zzzzz...
Request someone to please assist me asap.

Replies: 9 | **Pages:** 1 - **Last Post:** Mar 23, 2013 2:12 PM by: Joe@AWS

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Replies

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: cmpe295

Posted on: Mar 22, 2013 10:47 AM

in response to: cmpe295

Hello,

Can someone from the AWS team please look into this?

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: cmpe295

Posted on: Mar 22, 2013 12:12 PM

in response to: cmpe295

These are the instance ids for the instances for your reference -

i-59d6b236
i-6a60ad01

I have hadoop master running on i-59d6b236, and hadoop slave configured on i-6a60ad01. I am able to ssh from one instance to the other without any issues. But when I start the HDFS layer on the master using the command - bin/start-dfs.sh, I see the errors in the slave log.

hadoop@slave$ tail -10 hadoop-hadoop-datanode-ip-10-245-218-252.log
2013-03-22 19:11:45,735 INFO org.apache.hadoop.ipc.Client: Retrying connect to server: master/54.225.206.51:9000. Already tried 7 time(s).
2013-03-22 19:11:46,737 INFO org.apache.hadoop.ipc.Client: Retrying connect to server: master/54.225.206.51:9000. Already tried 8 time(s).
2013-03-22 19:11:47,739 INFO org.apache.hadoop.ipc.Client: Retrying connect to server: master/54.225.206.51:9000. Already tried 9 time(s).
2013-03-22 19:11:47,741 INFO org.apache.hadoop.ipc.RPC: Server at **master/54.225.206.51:9000 not available yet, Zzzzz...**
2013-03-22 19:11:49,743 INFO org.apache.hadoop.ipc.Client: Retrying connect to server: master/54.225.206.51:9000. Already tried 0 time(s).
2013-03-22 19:11:50,745 INFO org.apache.hadoop.ipc.Client: Retrying connect to server: master/54.225.206.51:9000. Already tried 1 time(s).
2013-03-22 19:11:51,747 INFO org.apache.hadoop.ipc.Client: Retrying connect to server:

master/54.225.206.51:9000. Already tried 2 time(s).
2013-03-22 19:11:52,749 INFO org.apache.hadoop.ipc.Client: **Retrying connect to server:**
master/54.225.206.51:9000. Already tried 3 time(s).
2013-03-22 19:11:53,751 INFO org.apache.hadoop.ipc.Client: Retrying connect to server:
master/54.225.206.51:9000. Already tried 4 time(s).
2013-03-22 19:11:54,752 INFO org.apache.hadoop.ipc.Client: Retrying connect to server:
master/54.225.206.51:9000. Already tried 5 time(s).

Request someone to please look into it asap.

Edited by: cmpe295 on Mar 22, 2013 12:12 PM

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: cmpe295

Posted on: Mar 22, 2013 12:16 PM

in response to: cmpe295

To give you some additional info -

1. I have defined the elastic IPs for my master and slave, in the /etc/hosts file on both the master and the slave node.

hadoop@slave$ cat /etc/hosts
54.225.206.51 master
54.225.207.96 slave

2. I have opened the ports 9000, 9001, 50000-50100 on my default security group.
3. I am able to ssh from master-master, slave-slave, master-slave, and slave-master without any issues.

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: Joe@AWS

Posted on: Mar 22, 2013 1:19 PM

in response to: cmpe295

Hi cmpe295,

I have taken a look at your instances and see that the security groups appear to be configured properly to allow traffic from the world (0.0.0.0/0) to the ports you've listed, which is great.

Here are some steps you can take to determine what the problem might be:

• Confirm there is a Hadoop process running on the master accepting TCP connections

For example, this is showing that the SSH process is listening on my SSH server:

```
$ sudo netstat -antp |grep ':22' |grep LISTEN
tcp        0      0 0.0.0.0:22                  0.0.0.0:*                   LISTEN
1100/sshd
tcp        0      0 127.0.0.1:2200              0.0.0.0:*                   LISTEN
1113/sshd
tcp        0      0 :::22                       :::*                        LISTEN
1100/sshd
tcp        0      0 ::1:2200                    :::*                        LISTEN
1113/sshd
```

If you don't see anything when you run:

```
sudo netstat -antp |grep '9000'
```

...something may be wrong with Hadoop (perhaps it is not running properly on the Master node).

• Verify that network configuration by seeing if traffic is reaching the master instance by performing a packet capture:

Start packet capture on master node:

```
master $ sudo tcpdump tcp port 9000
```

Attempt to establish TCP connection on port 9000 with the master from the slave (even if nothing is listening on the master node, you should see network traffic showing up on the master node if networking is configured properly):

```
slave $ telnet 54.225.206.51 9000
```

```
Trying 54.225.206.51...
telnet: connect to address 54.225.206.51: Connection refused
telnet: Unable to connect to remote host
```

If the networking is configured properly, you will see something like this:

```
master $ sudo tcpdump tcp port 9000
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 65535 bytes
20:13:26.182392 IP ec2-X-X-X-X.us-west-2.compute.amazonaws.com.49774 >
10.0.0.222.cslistener: Flags [S], seq 4033130444, win 8192, options [mss 1460,nop,wscale
8,nop,nop,sackOK], length 0
20:13:26.182425 IP 10.0.0.222.cslistener > ec2-X-X-X-X.us-west-
2.compute.amazonaws.com.49774: Flags [R.], seq 0, ack 4033130445, win 0, length 0
20:13:26.679246 IP ec2-X-X-X-X.us-west-2.compute.amazonaws.com.49774 >
10.0.0.222.cslistener: Flags [S], seq 4033130444, win 8192, options [mss 1460,nop,wscale
8,nop,nop,sackOK], length 0
20:13:26.679258 IP 10.0.0.222.cslistener > ec2-X-X-X-X.us-west-
2.compute.amazonaws.com.49774: Flags [R.], seq 0, ack 1, win 0, length 0
20:13:27.179507 IP ec2-X-X-X-X.us-west-2.compute.amazonaws.com.49774 >
10.0.0.222.cslistener: Flags [S], seq 4033130444, win 8192, options [mss
1460,nop,nop,sackOK], length 0
20:13:27.179519 IP 10.0.0.222.cslistener > ec2-X-X-X-X.us-west-
2.compute.amazonaws.com.49774: Flags [R.], seq 0, ack 1, win 0, length 0
```

in which case, it is likely that there is something wrong with the Hadoop process, rather than a networking issue.

I hope this helps!

Regards,
Joe C.

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: cmpe295

Posted on: Mar 22, 2013 1:56 PM

in response to: Joe@AWS

Thanks Joe. I checked for the process on the master node -

root@master# netstat -antp |grep ':22' |grep LISTEN
tcp 0 0 0.0.0.0:22 0.0.0.0:* LISTEN 683/sshd
tcp6 0 0 :::22 :::* LISTEN 683/sshd
root@master# sudo netstat -antp |grep '9000'
root@master#

You are right. There is no hadoop process running. I checked the logs on the master node this time to see why its failing, and the log file is filled with the below messages

2013-03-22 17:43:53,802 ERROR org.apache.hadoop.hdfs.server.namenode.NameNode: java.net.BindException: Problem binding to master/54.225.206.51:9000 : Cannot assign requested address
at org.apache.hadoop.ipc.Server.bind(Server.java:227)
at org.apache.hadoop.ipc.Server$Listener.<init>(Server.java:301)
at org.apache.hadoop.ipc.Server.<init>(Server.java:1483)
at org.apache.hadoop.ipc.RPC$Server.<init>(RPC.java:545)
at org.apache.hadoop.ipc.RPC.getServer(RPC.java:506)
at org.apache.hadoop.hdfs.server.namenode.NameNode.initialize(NameNode.java:294)
at org.apache.hadoop.hdfs.server.namenode.NameNode.<init>(NameNode.java:496)
at org.apache.hadoop.hdfs.server.namenode.NameNode.createNameNode(NameNode.java:1279)
at org.apache.hadoop.hdfs.server.namenode.NameNode.main(NameNode.java:1288)
Caused by: java.net.BindException: Cannot assign requested address

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: cmpe295

Posted on: Mar 22, 2013 2:40 PM

in response to: cmpe295

Joe,

Can you please look into this?

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: Joe@AWS

Posted on: Mar 22, 2013 3:00 PM

in response to: cmpe295

Hi,

Looking at your output above:

```
hadoop@slave$ cat /etc/hosts
54.225.206.51 master
54.225.207.96 slave
```

and doing a quick search online, it looks like you may be running into an issue similar to the one outlined in this StackOverflow thread.

I think your modifications to the /etc/hosts file may be causing the issue. Generally, it is not a good idea to remove lines like this from your /etc/hosts file:

```
127.0.0.1   localhost localhost.localdomain
```

Please try to add the line back to see if it will fix the Hadoop configuration issues inside your instance.

I hope this helps!

Regards,
Joe C.

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: cmpe295

Posted on: Mar 22, 2013 3:25 PM

in response to: Joe@AWS

I haven't commented it. I just pasted the last two lines from my /etc/hosts file. Here is the complete entry -

root@master# cat /etc/hosts
127.0.0.1 localhost

1. The following lines are desirable for IPv6 capable hosts

::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts

54.225.206.51 master
54.225.207.96 slave

I have a similar entry on the slave instance too.

Re: Unable to communicate between hadoop master to Slave ec2 instances

Reply

Posted by: Joe@AWS

Posted on: Mar 23, 2013 2:11 PM

in response to: cmpe295

Hi everyone,

I just wanted to paste the solution here so the thread isn't left open ended. The issue was caused by an inappropriate entry in the /etc/hosts file. In this case, it was:

```
54.225.206.51 master
```

> *An EC2 instance does not actually have an external IP address registered in its interface list in Linux, and you are trying to change the addressing to an external IP address, which is causing hadoop to try to bind port 9000 on address 54.225.206.51 which doesn't actually exist in your Linux instance, so it causes an error:*
>
> ```
> 2013-03-22 17:43:53,802 ERROR org.apache.hadoop.hdfs.server.namenode.NameNode:
> java.net.BindException: Problem binding to master/54.225.206.51:9000 : Cannot assign
> ```

```
requested address
```

I hope this helps!

Regards,
Joe C.

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